



SupportOps

A multi-channel customer support platform. The AI Agent handles conversations first, and hands them to a Human Agent once it is no longer safe to continue.

- Web Widget
- WhatsApp
- RAG + MCP Tools
- Escalation to Human Agent
- Evaluated & measured

Three separate inboxes

Every Channel stands alone: its own app, its own account, its own history.



Website live chat

Answered by whoever happens to notice



WhatsApp

Run from one staff member's personal phone



Email

Threads that never truly get closed

no shared context, queue, or knowledge base

The impact

The cost is borne every day by the smallest team in the company.



Slow response time

No clear owner per Channel



Broken context

One Customer, three separate transcripts



Duplicated work

The same answer written twice



Scattered knowledge

Policies live in documents rarely opened



Repetitive questions

The same topics asked every day



Inefficient handoff

The next agent has to reread the whole conversation

3

separate apps to serve one Customer

Rp3.75-5.48M

monthly salary range for Customer Service in Indonesia

24/7

questions can arrive outside the team's working hours

1

team has to cover every Channel and its context

Approach	Answers from the company's own knowledge	Accesses business data in real time	Still holds when a new Channel is added	Cost model
Hire more Human Agents The most common solution	✓	✓	~ one more inbox to watch	✗ monthly salary + operational capacity
Off-the-shelf AI support Example: Intercom Fin	✓ can use the company help center	~ depends on integrations and plan	✓ chat and email supported	~ from US\$0.99 per outcome + seats
Web Widget-only AI AI built directly into the web channel	~ possible, but has to be built	~ integrations built only for web	✗ Channel logic is baked into the AI	~ cheap model, but development cost
SupportOps AI-first, human on escalation	✓ RAG over Customer-Safe Knowledge Sources	✓ HTTP- and MCP-based Business Tools	✓ decoupled by Channel Adapters	✓ ±Rp113 per typical Session, model cost

The need is not just adding AI, but answers that are grounded in sources, scoped per Workspace, and Channel-independent from day one.



Already live

The complete support flow, from the first message until the Ticket is closed or handed to a Human Agent.



Channels & conversations

Web Widget · WhatsApp · Pre-Chat · Session Link · streaming via SSE



AI Agent capabilities

RAG · Ticket history · attachments · HTTP & MCP Tool read/write



Human Agent side

Escalation · Shared Queue · Claim · Takeover · AI Copilot



Operations

Follow-Up · Auto-Resolution · Activity Timeline · analytics dashboard

Out of MVP scope

Email as a Channel

Role Supervisor

SLA engine

Auto-routing

Knowledge versioning

Scheduled recrawl

Cross-channel identity

SSO

Multi-workspace user

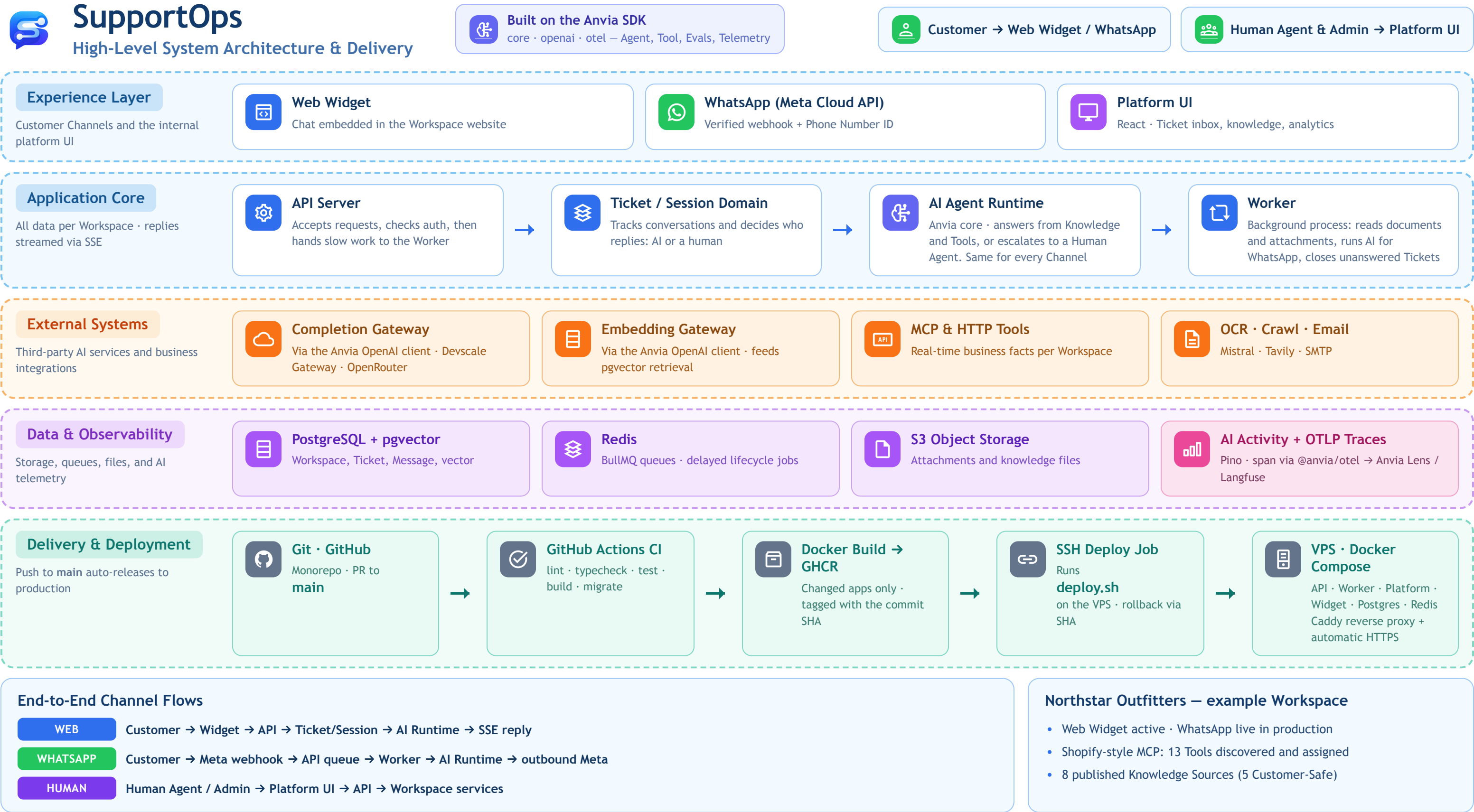
Fallback AI provider

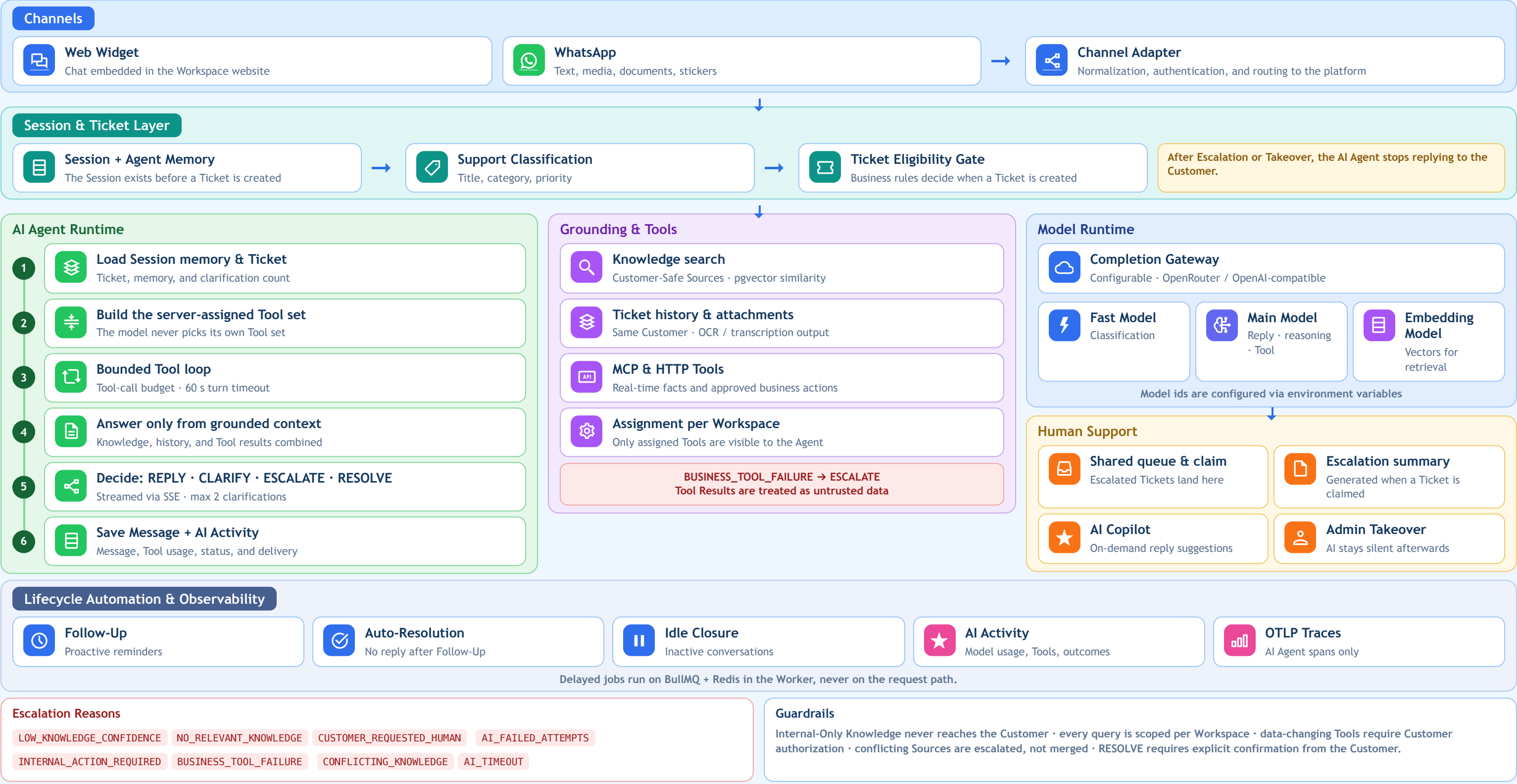
Multi-model

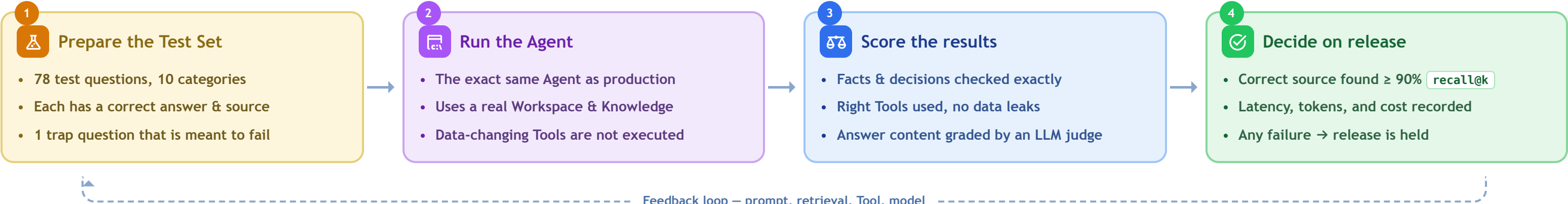
Ticket reopen

Conversation search

These items are deliberately deferred so the MVP proves the core support flow first.









Coverage of 78 Cases

22 common — policies & facts

11 edge — corner cases

9 toolCall — Shopify lookups

9 escalation — hand off to a human

6 staleness — conflicting sources

5 abstention — refuses to guess

4 visibility — no internal leaks

4 guardrail — injection & card data

4 language — ID / EN

4 attachment — PDF, image, voice



Models in the run



Main — under test

gpt-5.6-luna



Fast — classification

deepseek-v4-flash-0731



Embedding — retrieval

text-embedding-3-small



Judge — scoring only

gemini-3.8-flash · 0.7



How the Agent got to pass

- Freeze the baseline first — `B0 → B3`
- One change, then rerun the suite
- Fix the prompt, not the test
- The quality floor must never drop
- Tune retrieval on recall, not gut feeling
- Cheaper only if quality holds — Tool manifest -76%

B0 → B3

every scored suite now passes, and nothing that passed before has regressed



Results — all suites, B0 → B3

Suite	Case	B0 16 Sep	B3 17 Sep
g-eval — answer quality	23	73.9%	100%
decision — REPLY / CLARIFY / ESCALATE / RESOLVE	14	85.7%	100%
tool usage — the right Tool, no forbidden calls	7	85.7%	100%
faithfulness — grounded in retrieved Chunks	3	50.0%	100%
retriever / retrieval — recall & ranking	10 + 10	—	100%
contains · exact match — exact facts	6 + 2	100%	100%
visibility — no Internal-Only leaks	6	100%	100%
language · relevancy	3 + 3	100%	100%
negative control — must fail	1	0%	0%

Read in Anvia Lens: every Case is a score with its input, answer, tokens, and latency — through the same OTLP pipeline as the Agent's own traces.

LOW

Mostly FAQ and small talk

\$39

/ month

Rp698,284 · Rp70 per Session

Main LLM

OCR

AI Copilot

Other

\$37.36

\$0.00

\$1.00

\$1.08

MEDIUM — the mix we use as our baseline

FAQ + Shopify lookups + some purchases

\$64

/ month

Rp1,130,442 · Rp113 per Session

Main LLM

OCR

AI Copilot

Other

\$57.80

\$4.00

\$1.00

\$1.06

HIGH

Commerce-heavy, long conversations

\$89

/ month

Rp1,575,071 · Rp158 per Session

Main LLM

OCR

AI Copilot

Other

\$78.86

\$8.00

\$1.00

\$1.12

🗨️

Cost of one Session

median of 3 measured runs

small talk		\$0.0001
one FAQ		\$0.0014
escalation		\$0.0027
order lookup		\$0.0091
five FAQs		\$0.0108
purchase		\$0.0112
attachment		\$0.0119
15 turns		\$0.0149

The main model makes up ~90% of the bill · 77-96% of its input is cached, 10× cheaper per token · OCR only appears with attachments

📊

If volume grows

SESSIONS / MONTH	LOW	MEDIUM	HIGH
1,000	\$4 Rp78K	\$7 Rp121K	\$9 Rp165K
10,000	\$39 Rp698K	\$64 Rp1.13M	\$89 Rp1.58M
100,000	\$390 Rp6.9M	\$634 Rp11.2M	\$886 Rp15.7M

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One conversation vs the market

log scale

SupportOps

Intercom Fin

Zendesk AI

Human Agent

Rp113

Rp17K

Rp26–35K

Rp18–120K

~100-300× cheaper — but our figure is raw model cost, while theirs is retail price. Add server, database, and storage (\$20-40 / month) and 10,000 Sessions still stay under \$100.



DEMO

One conversation, from start to finish



Week 1 · Hardening

Closing the technical risks already found in the MVP.



Customer Identity to Tool

Demo Tools still answer from one fixed account



Eval as a CI gate

Still run manually before each release



AI Usage & Cost Analytics

Token and cost dashboard per AI Turn, Session, and Workspace



Durable retry

Widget message retries live in API memory, lost on restart



Scheduled discovery

MCP discovery and knowledge recrawl are still manual

Weeks 2-3 · Pilot

Testing real flows with Customers and Human Agents, not just Eval Cases.



Two real Channels

Web Widget and WhatsApp on one pilot Workspace



FAQ & policy

Measure grounded answers and no-knowledge cases



Business lookups

Orders, invoices, products, and inventory from the Customer's account



Mutating actions

Cart updates with the proper request and confirmation



Success criteria

Resolution, Escalation, latency, error rate, and cost per Session

Weeks 4-6 · Scale-up

Expanding Channels, team operations, and platform resilience based on pilot results.



Email as the third Channel

Its adapter pattern is already proven twice



Cross-channel identity

Linking the same Customer across Channels



Supervisor, SLA, auto-routing

For large teams, not a single operator



Knowledge versioning

Roll back when a publish goes wrong



Fallback AI provider

Model selection per Workspace



Thank you

Any questions?